Ising Model Notes: From Partition Function to Magnetization

Chittaranja Hens (Structured Version)

1 Why the Partition Function?

Core Idea

The partition function is the central object of statistical mechanics:

$$Z = \sum_{\{\sigma\}} e^{-\beta H}$$

It contains **all thermodynamic information**.

From Z we get:

$$F = -k_B T \ln Z \tag{1}$$

$$E = -\frac{\partial \ln Z}{\partial \beta} \tag{2}$$

$$M = \frac{\partial \ln Z}{\partial(\beta H)} \tag{3}$$

$$C = \frac{\partial E}{\partial T} \tag{4}$$

Physical Meaning

- Low temperature: lowest energy states dominate
- High temperature: many configurations contribute

2 1D Ising Model (No Field, No PBC)

2.1 Hamiltonian

$$H = -J \sum_{i=1}^{N-1} \sigma_i \sigma_{i+1} \tag{5}$$

2.2 Physical Picture

Open chain → edge spins behave differently (no symmetry).

3 Explicit Partition Function Construction

3.1 Explicit Construction for Small Systems

We begin by evaluating the partition function for small system sizes in order to identify a pattern.

3.1.1 Two Spins

$$Z_2 = \sum_{\sigma_1,\sigma_2} e^{\beta J \sigma_1 \sigma_2} \tag{6}$$
$$= e^{\beta J} + e^{\beta J} + e^{-\beta J} + e^{-\beta J} \tag{7}$$
$$= 4\cosh(\beta J) \tag{8}$$

3.1.2 Three Spins

$$Z_3 = \sum_{\sigma_1,\sigma_2,\sigma_3} e^{\beta J(\sigma_1\sigma_2 + \sigma_2\sigma_3)} \tag{9}$$
$$= 2(e^{2\beta J} + e^{-2\beta J}) + 4 \tag{10}$$
$$= 2(2\cosh \beta J)^2 \tag{11}$$

3.1.3 General Form

$$Z_N = 2(2\cosh \beta J)^{N-1} \tag{12}$$

3.2 General Result

Important Insight

Factorization occurs because the chain has no loop.

4 Thermodynamics (No Field)

$$F = -Nk_BT \ln(2\cosh \beta J) \tag{13}$$
$$E = -NJ \tanh(\beta J) \tag{14}$$
$$C = Nk_B(\beta J)^2 \operatorname{sech}^2(\beta J) \tag{15}$$

5 Thermodynamics (No Field)

Starting from:

$$Z_N = (2\cosh \beta J)^N \tag{16}$$

5.1 Free Energy

$$\ln Z_N = N \ln(2\cosh\beta J) \tag{17}$$

$$F = -k_B T \ln Z_N \tag{18}$$

$$\boxed{F = -Nk_B T \ln(2\cosh\beta J)} \tag{19}$$

5.2 Energy

$$E = -\frac{\partial \ln Z}{\partial \beta} \tag{20}$$

$$\begin{aligned}
\frac{\partial \ln Z}{\partial \beta} &= N\frac{1}{2\cosh\beta J} \cdot 2\sinh\beta J \cdot J && (21)\\
&= NJ\frac{\sinh\beta J}{\cosh\beta J} && (22)\\
&= NJ\tanh(\beta J) && (23)
\end{aligned}$$

$$\boxed{E = -NJ\tanh(\beta J)} \tag{24}$$

5.3 Heat Capacity

$$C = k_B\beta^2 \frac{\partial^2 \ln Z}{\partial \beta^2} \tag{25}$$

$$\begin{aligned}
\frac{\partial^2 \ln Z}{\partial \beta^2} &= NJ\frac{d}{d\beta}\tanh(\beta J) && (26)\\
&= NJ \cdot J\,\mathrm{sech}^2(\beta J) && (27)
\end{aligned}$$

$$\boxed{C = Nk_B(\beta J)^2\,\mathrm{sech}^2(\beta J)} \tag{28}$$

6 Introducing Periodic Boundary Conditions (PBC)

With periodic boundary condition:

$$\sigma_{N+1} = \sigma_1 \tag{29}$$

The Hamiltonian becomes:

$$H = -J\sum_{i=1}^{N} \sigma_i\sigma_{i+1} \tag{30}$$

Now the system forms a closed loop, and the previous factorization method fails.

Now all spins are equivalent $\rightarrow$ translational symmetry.

6.1 Effect on Partition Function

Now interaction includes:

$$\sigma_N \sigma_1$$

This breaks simple factorization → need new method.

This is where the **transfer matrix** becomes essential.

7 Add External Magnetic Field

7.1 Hamiltonian

$$H = -J \sum_{i=1}^{N} \sigma_i \sigma_{i+1} - H \sum_{i=1}^{N} \sigma_i \tag{31}$$

Now spins feel:

- Interaction (neighbor coupling)
- External bias (field H)

7.2 Partition Function as Matrix Product

$$Z_N = \mathrm{Tr}(\tilde{T}^N) \tag{32}$$

Summing over spins = matrix multiplication.

8 Add External Magnetic Field

$$H = -J \sum_{i=1}^{N} \sigma_i \sigma_{i+1} - H \sum_{i=1}^{N} \sigma_i \tag{33}$$

$$Z_N = \sum_{\{\sigma\}} \exp\left[\beta J \sum \sigma_i \sigma_{i+1} + \beta H \sum \sigma_i\right] \tag{34}$$

9 Transfer Matrix Construction (Detailed)

9.1 Step 1: Rewrite Partition Function

$$Z_N = \sum_{\{\sigma\}} \prod_{i=1}^{N} \exp\left[\beta J \sigma_i \sigma_{i+1} + \beta H \sigma_i\right] \tag{35}$$

9.2 Step 2: Symmetrization

To make interaction symmetric:

$$\beta H\sigma_i = \frac{\beta H}{2}\sigma_i + \frac{\beta H}{2}\sigma_i \tag{36}$$

Thus:

$$Z_N = \sum_{\{\sigma\}} \prod_{i=1}^{N} \exp\left[\beta J\sigma_i\sigma_{i+1} + \frac{\beta H}{2}(\sigma_i + \sigma_{i+1})\right] \tag{37}$$

9.3 Step 3: Define Transfer Matrix

$$\tilde{T}_{\sigma_i,\sigma_{i+1}} = \exp\left[\beta J\sigma_i\sigma_{i+1} + \frac{\beta H}{2}(\sigma_i + \sigma_{i+1})\right] \tag{38}$$

9.4 Step 4: Explicit Matrix Elements

Since $\sigma = \pm 1$:

$$\tilde{T}_{++} = e^{\beta(J+H)} \tag{39}$$
$$\tilde{T}_{+-} = e^{-\beta J} \tag{40}$$
$$\tilde{T}_{-+} = e^{-\beta J} \tag{41}$$
$$\tilde{T}_{--} = e^{\beta(J-H)} \tag{42}$$

$$\tilde{T} = \begin{pmatrix} e^{\beta(J+H)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} \end{pmatrix} \tag{43}$$

9.5 Step 5: Matrix Multiplication Structure

$$Z_N = \sum_{\sigma_1,\ldots,\sigma_N} \tilde{T}_{\sigma_1,\sigma_2}\tilde{T}_{\sigma_2,\sigma_3}\cdots\tilde{T}_{\sigma_N,\sigma_1} \tag{44}$$

Now summing over intermediate spins:

$$\sum_{\sigma_2} \tilde{T}_{\sigma_1,\sigma_2}\tilde{T}_{\sigma_2,\sigma_3} = (\tilde{T}^2)_{\sigma_1,\sigma_3} \tag{45}$$

Repeating:

$$Z_N = \sum_{\sigma_1} (\tilde{T}^N)_{\sigma_1,\sigma_1} \tag{46}$$

$$\boxed{Z_N = \mathrm{Tr}(\tilde{T}^N)} \tag{47}$$

9.6 Linear Algebra Interpretation of $\mathrm{Tr}(\tilde{T}^N)$

We have obtained:

$$Z_N = \mathrm{Tr}(\tilde{T}^N) \tag{48}$$

To understand why this is useful, we diagonalize the transfer matrix.

9.7 Diagonalization

Since $\tilde{T}$ is a real symmetric matrix, it can be diagonalized:

$$\tilde{T} = UDU^{-1} \tag{49}$$

where:

- U is a matrix of eigenvectors
- D is diagonal:

$$D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}$$

9.8 Power of the Matrix

Now consider $\tilde{T}^N$:

$$\tilde{T}^N = (UDU^{-1})(UDU^{-1}) \cdots (UDU^{-1}) \tag{50}$$

Using $U^{-1}U = I$, all middle terms cancel:

$$\tilde{T}^N = UD^N U^{-1} \tag{51}$$

9.9 Why D^N is Simple

Since D is diagonal:

$$D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix} \tag{52}$$

Then:

$$D^N = \begin{pmatrix} \lambda_+^N & 0 \\ 0 & \lambda_-^N \end{pmatrix} \tag{53}$$

Key Linear Algebra Fact

Raising a diagonal matrix to a power simply raises each eigenvalue to that power.

9.10 Trace is Invariant

Now compute the partition function:

$$Z_N = \mathrm{Tr}(\tilde{T}^N) \tag{54}$$
$$= \mathrm{Tr}(UD^N U^{-1}) \tag{55}$$

Using cyclic property of trace:

$$\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) \tag{56}$$

we get:

$$Z_N = \mathrm{Tr}(D^N) \tag{57}$$

9.11 Final Result

$$\boxed{Z_N = \lambda_+^N + \lambda_-^N} \tag{58}$$

9.12 Physical Interpretation

- Each eigenvalue represents a possible "mode" of the system
- Raising to power N corresponds to propagating through N spins
- The largest eigenvalue dominates as $N \to \infty$

$$Z_N \approx \lambda_+^N \quad (N \to \infty) \tag{59}$$

10 Eigenvalues of the Transfer Matrix (Full Derivation)

Solve:

$$\det\left(\tilde{T} - \lambda I\right) = 0 \tag{60}$$

$$\lambda_\pm = e^{\beta J}\left[\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}}\right] \tag{61}$$

Thus:

$$Z_N = \lambda_+^N + \lambda_-^N \tag{62}$$

The transfer matrix is:

$$\tilde{T} = \begin{pmatrix} e^{\beta(J+H)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} \end{pmatrix} \tag{63}$$

10.1 Step 1: Characteristic Equation

Eigenvalues are obtained from:

$$\det\left(\tilde{T} - \lambda I\right) = 0 \tag{64}$$

$$\begin{vmatrix} e^{\beta(J+H)} - \lambda & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} - \lambda \end{vmatrix} = 0 \tag{65}$$

10.2 Step 2: Expand Determinant

$$\left(e^{\beta(J+H)} - \lambda\right)\left(e^{\beta(J-H)} - \lambda\right) - e^{-2\beta J} = 0 \tag{66}$$

10.3 Step 3: Multiply Terms

$$e^{\beta(J+H)}e^{\beta(J-H)} - \lambda e^{\beta(J+H)} - \lambda e^{\beta(J-H)} + \lambda^2 - e^{-2\beta J} = 0 \tag{67}$$

10.4 Step 4: Simplify Exponentials

First term:

$$e^{\beta(J+H)}e^{\beta(J-H)} = e^{2\beta J} \tag{68}$$

So we get:

$$\lambda^2 - \lambda\left(e^{\beta(J+H)} + e^{\beta(J-H)}\right) + e^{2\beta J} - e^{-2\beta J} = 0 \tag{69}$$

10.5 Step 5: Simplify Coefficients

Use identities:

$$e^{\beta(J+H)} + e^{\beta(J-H)} = e^{\beta J}(e^{\beta H} + e^{-\beta H}) \tag{70}$$

$$= 2e^{\beta J}\cosh(\beta H) \tag{71}$$

Also:

$$e^{2\beta J} - e^{-2\beta J} = (e^{\beta J})^2 - (e^{-\beta J})^2 \tag{72}$$

10.6 Step 6: Quadratic Equation

$$\lambda^2 - 2e^{\beta J}\cosh(\beta H)\lambda + \left(e^{2\beta J} - e^{-2\beta J}\right) = 0 \tag{73}$$

10.7 Step 7: Solve Quadratic

Using:

$$\lambda = \frac{a \pm \sqrt{a^2 - 4b}}{2}$$

we identify:

$$a = 2e^{\beta J}\cosh(\beta H), \quad b = e^{2\beta J} - e^{-2\beta J}$$

Thus:

$$\lambda_\pm = \frac{2e^{\beta J}\cosh(\beta H) \pm \sqrt{(2e^{\beta J}\cosh(\beta H))^2 - 4(e^{2\beta J} - e^{-2\beta J})}}{2} \tag{74}$$

10.8 Step 8: Simplify Square Root

$$(2e^{\beta J}\cosh(\beta H))^2 = 4e^{2\beta J}\cosh^2(\beta H) \tag{75}$$

So inside square root:

$$4e^{2\beta J}\cosh^2(\beta H) - 4(e^{2\beta J} - e^{-2\beta J}) \tag{76}$$

Factor 4:

$$= 4\left[e^{2\beta J}\cosh^2(\beta H) - e^{2\beta J} + e^{-2\beta J}\right] \tag{77}$$

10.9 Step 9: Use Identity

$$\cosh^2 x - 1 = \sinh^2 x \tag{78}$$

So:

$$e^{2\beta J}(\cosh^2(\beta H) - 1) = e^{2\beta J}\sinh^2(\beta H) \tag{79}$$

Thus:

$$= 4\left[e^{2\beta J}\sinh^2(\beta H) + e^{-2\beta J}\right] \tag{80}$$

10.10 Step 10: Final Expression

Taking square root:

$$\sqrt{4\left[e^{2\beta J}\sinh^2(\beta H) + e^{-2\beta J}\right]} = 2e^{\beta J}\sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \tag{81}$$

10.11 Final Result

$$\boxed{\lambda_\pm = e^{\beta J}\left[\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}}\right]} \tag{82}$$

11 Magnetization (Detailed Derivation)

11.1 From Partition Function

Differentiate Z:

$$\boxed{m = \frac{1}{N}\frac{\partial \ln Z}{\partial(\beta H)}} \tag{83}$$

11.2 Thermodynamic Limit

Since $\lambda_+ > \lambda_-$:

$$Z \approx \lambda_+^N \tag{84}$$

$$\ln Z = N \ln \lambda_+ \tag{85}$$

$$\begin{aligned} m &= \frac{1}{N}\frac{\partial \ln Z}{\partial(\beta H)} && (86)\\ &= \frac{\partial \ln \lambda_+}{\partial(\beta H)} && (87)\end{aligned}$$

$$\boxed{m = \frac{\partial \ln \lambda_+}{\partial(\beta H)}} \tag{88}$$

Key Result: No phase transition in 1D.

12 Magnetization from the Largest Eigenvalue

12.1 Step 1: Starting Point

In the thermodynamic limit:

$$Z \approx \lambda_+^N \tag{89}$$

Thus:

$$\ln Z = N \ln \lambda_+ \tag{90}$$

Magnetization per spin is:

$$m = \frac{1}{N} \frac{\partial \ln Z}{\partial(\beta H)} \tag{91}$$

Substituting:

$$m = \frac{1}{N} \frac{\partial}{\partial(\beta H)} (N \ln \lambda_+) \tag{92}$$

$$= \frac{\partial \ln \lambda_+}{\partial(\beta H)} \tag{93}$$

12.2 Step 2: Expression for λ_+

$$\lambda_+ = e^{\beta J} \left[\cosh(\beta H) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right] \tag{94}$$

Define:

$$f(H) = \cosh(\beta H) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \tag{95}$$

Thus:

$$\lambda_+ = e^{\beta J} f(H) \tag{96}$$

12.3 Step 3: Take Logarithm

$$\ln \lambda_+ = \beta J + \ln f(H) \tag{97}$$

Since βJ does not depend on H:

$$m = \frac{\partial \ln f(H)}{\partial(\beta H)} \tag{98}$$

12.4 Step 4: Differentiate $f(H)$

Let:

$$f(H) = \cosh(\beta H) + g(H) \tag{99}$$

where:

$$g(H) = \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \tag{100}$$

Now differentiate:

$$\frac{df}{d(\beta H)} = \sinh(\beta H) + \frac{dg}{d(\beta H)} \tag{101}$$

12.5 Step 5: Differentiate Square Root Term

$$\frac{dg}{d(\beta H)} = \frac{1}{2\sqrt{\sinh^2(\beta H) + e^{-4\beta J}}} \cdot \frac{d}{d(\beta H)} \left(\sinh^2(\beta H)\right) \tag{102}$$

Now:

$$\frac{d}{d(\beta H)} \left(\sinh^2(\beta H)\right) = 2\sinh(\beta H)\cosh(\beta H) \tag{103}$$

Thus:

$$\frac{dg}{d(\beta H)} = \frac{\sinh(\beta H)\cosh(\beta H)}{\sqrt{\sinh^2(\beta H) + e^{-4\beta J}}} \tag{104}$$

12.6 Step 6: Combine Results

$$\frac{df}{d(\beta H)} = \sinh(\beta H) + \frac{\sinh(\beta H)\cosh(\beta H)}{\sqrt{\sinh^2(\beta H) + e^{-4\beta J}}} \tag{105}$$

12.7 Step 7: Final Expression for Magnetization

$$m = \frac{1}{f(H)} \frac{df}{d(\beta H)} \tag{106}$$

Substituting:

$$\boxed{m = \frac{\sinh(\beta H) + \dfrac{\sinh(\beta H)\cosh(\beta H)}{\sqrt{\sinh^2(\beta H) + e^{-4\beta J}}}}{\cosh(\beta H) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}}}} \tag{107}$$

12.8 Step 8: Zero Field Limit

Let $H \to 0$:

$$\sinh(\beta H) \to 0 \tag{108}$$
$$\cosh(\beta H) \to 1 \tag{109}$$

Thus numerator $\to 0$, denominator finite:

$$\boxed{m = 0} \tag{110}$$

Conclusion

The 1D Ising model does not exhibit spontaneous magnetization.

12.9 Case 2: Small Field

- Spins align slightly with field
- Magnetization is smooth function of H

12.10 Case 3: Low Temperature

$$\beta J \gg 1 \Rightarrow m \approx 1 \tag{111}$$

12.11 Case 4: High Temperature

$$m \to 0 \tag{112}$$

13 Final Physical Picture

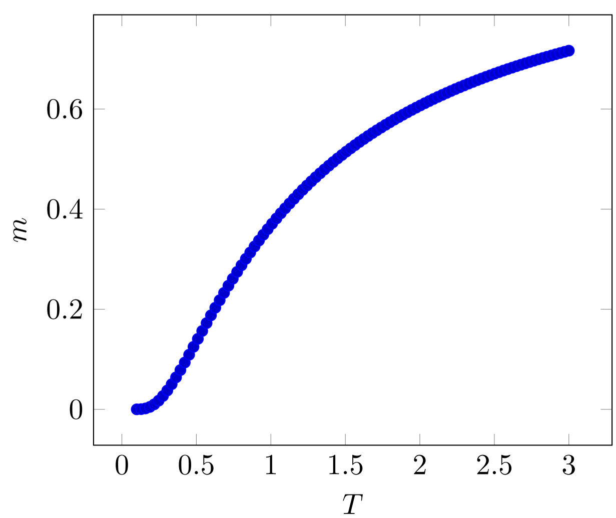
- 1D system cannot sustain long-range order
- Thermal fluctuations destroy alignment
- External field induces magnetization
- No phase transition

Big Takeaway

1D Ising Model: No spontaneous magnetization

14 Plots: Magnetization and Heat Capacity

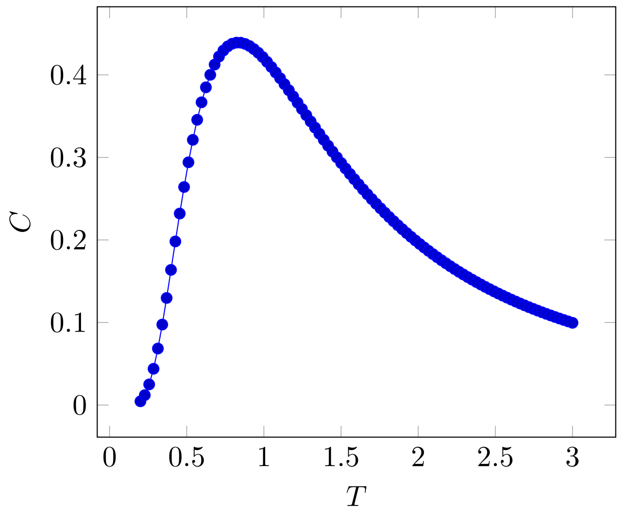
14.1 Magnetization vs Temperature (1D)



In 1D:

- Magnetization goes smoothly to zero
- No sharp transition

14.2 Heat Capacity



Heat capacity is smooth → no singularity → no phase transition.

15 Why No Phase Transition in 1D?

Physical Reason

Thermal fluctuations dominate in 1D:

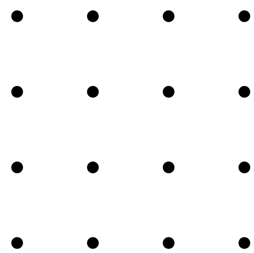
- Domain walls cost finite energy
- They appear easily at any $T > 0$
- Destroy long-range order

$$\uparrow \quad \uparrow \quad \downarrow \quad \downarrow \quad \uparrow$$

These domain walls break global alignment.

16 2D Ising Model (Contrast)

16.1 Key Difference



Each spin now interacts with 4 neighbors → stronger correlations.

16.2 Phase Transition

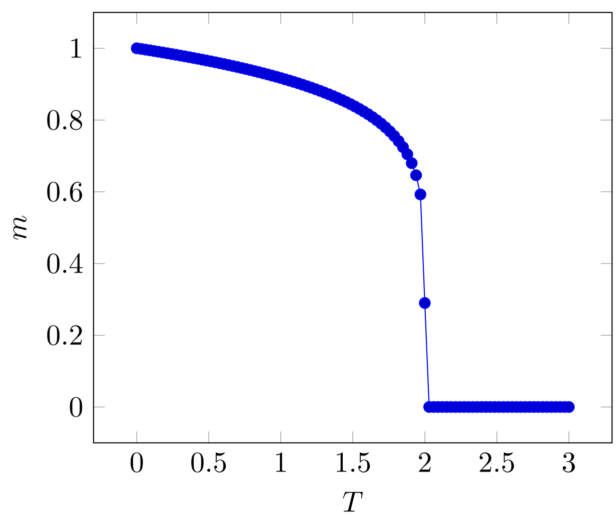
Critical temperature:

$$k_B T_c = \frac{2J}{\ln(1+\sqrt{2})} \tag{113}$$

Major Result

- $T < T_c$: ordered phase (non-zero magnetization)
- $T > T_c$: disordered phase

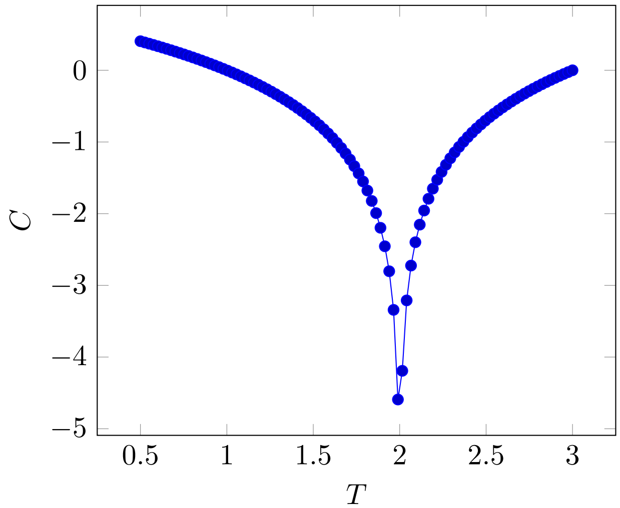
16.3 Magnetization Behavior (2D)



Unlike 1D:

- Magnetization drops sharply at T_c
- This is a true phase transition

16.4 Heat Capacity Divergence



Heat capacity diverges at $T_c \to$ signature of phase transition.

17 1D vs 2D Summary

	1D	2D
Phase Transition	No	Yes
Magnetization	Always 0 (H=0)	Non-zero below T_c
Heat Capacity	Smooth	Divergent

Final Insight

Dimensionality controls collective behavior.